

ABHIJAY KUMAR SHAH

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Education

Vellore Institute of Technology B.Tech in Computer Science & Engineering	Jul 2026 (Expected) CGPA: 8.44 / 10.0
DAV Public School, Dudhichua Senior Secondary (Class XII)	2019–2021 CGPA: 9.20 / 10.0
DAV Public School, Surya Vihar Secondary (Class X)	2018–2019 CGPA: 9.60 / 10.0

Technical Skills

Core Engineering: C++ (Advanced DSA), Python, SQL, **Object-Oriented Programming (OOP)**, DBMS, OS
Systems & Reliability: **Root Cause Analysis (RCA)**, Memory Management, System Scalability, Unit Testing (Jest)
Software Engineering: **SDLC (Requirement to Deployment)**, Git (Version Control), Docker, Technical Documentation
Full-Stack & Integration: Node.js, Express.js, React.js, RESTful APIs, JWT, RBAC, Cloud Infrastructure (Vercel/Render)
Automation & AI: Python-based Automation, Prompt Engineering, LLM Integration (Gemini/OpenRouter API)

Projects

- ProjectHub — Engineering Lifecycle & Compliance System** | *TypeScript, Node.js, PostgreSQL* [\[Link\]](#)
- Engineered a production-grade system lifecycle platform, managing technical workflows from requirement analysis to cloud deployment on Vercel and Render.
 - Implemented an automated Compliance Audit Engine using LLM-based structured JSON to validate technical documentation against predefined industrial standards.
 - Demonstrated Root Cause Analysis (RCA) by resolving critical production-level networking (IPv4/IPv6) and authentication bottlenecks, ensuring 100% system uptime.
 - Architected a secure REST API with 15+ endpoints, integrating role-based (RBAC) document workflows to ensure strict data integrity and process isolation.
- ParkEasy — Smart Urban Infrastructure & IoT Simulation** | *React, Node.js, MongoDB* [\[Link\]](#)
- Developed a Resource Optimization System for real-time parking slot allocation, focusing on minimizing search-time latency and improving urban infrastructure efficiency.
 - Designed robust back-end logic for state persistence and transactional integrity, ensuring zero-conflict booking in a high-concurrency concurrent environment.
 - Applied System Design principles to manage complex user sessions and secure data transmission using JWT and React Context API.
- Rately — Multi-Tenant Enterprise Quality Platform** | *React, Node.js, Security Hardening* [\[Link\]](#)
- Designed a scalable multi-tenant architecture with granular Role-Based Access Control (RBAC) to ensure secure data isolation across enterprise-tier users.
 - Implemented System Hardening using Rate Limiting and Helmet.js to mitigate DDoS threats, ensuring platform stability in an industrial production environment.
 - Focused on modular code architecture and clean separation of concerns to facilitate long-term system maintainability and scalability.

Experience / Leadership – Achievements

- CatCatchCode** [catcatchcode.online](#) | **Jul 2025 – Present**
- Co-Founder & Technical Lead – Ed-Tech Initiative [\[Link\]](#)
- Founded a technical initiative to bridge the industry-academia gap, providing mentorship on **C++**, **Data Structures**, and **System Design** to a community of 100+ students.
 - Led a cross-functional engineering team to produce 100+ modules on **SDLC best practices**, full-stack architecture, and AI-driven automation workflows.
 - Established standardized **version control (Git) workflows** and conducted rigorous code reviews to ensure high-quality, maintainable code across all community projects.
 - Designed comprehensive learning roadmaps focusing on **algorithmic efficiency and root-cause debugging**, preparing early-career engineers for core industry roles.
 - Leveraged performance analytics to monitor student engagement and content reach across YouTube and LinkedIn, driving **continuous process improvement**.

Certifications

- Python Essentials (VIT)** — May 2024 [\[Link\]](#)
- MERN Full Stack Certificate** — Apr 2025 [\[Link\]](#)